

SMART: A Surrogate Model for Predicting Application Runtime in Dragonfly Systems

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Why + What

Motivation

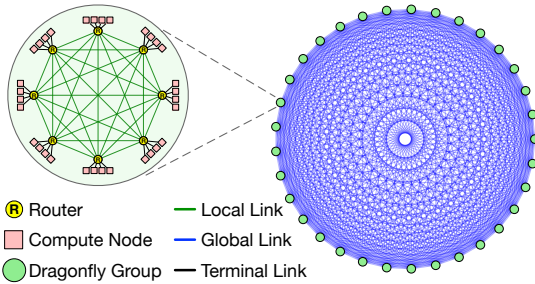
- Dragonfly is a dominant interconnect in modern HPC systems
- Application runtime is strongly affected by **network contention**
- High-fidelity PDES is accurate but **prohibitively slow**
- Fast and accurate **runtime prediction** is critical for hybrid simulation

Goal

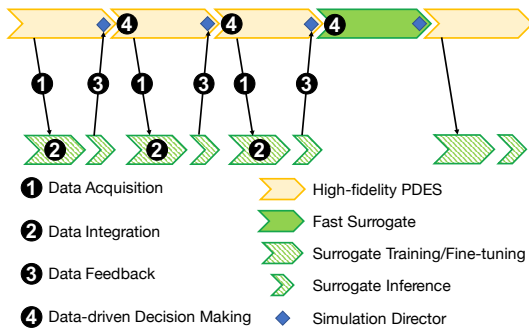
Predict the **next iteration runtime** of concurrent parallel applications on Dragonfly networks using **runtime history** and **network state**.

What is Dragonfly system?

Topology of a 1,056-node 1D Dragonfly network.



What is Hybrid PDES-Surrogate Simulation?



- Hybrid simulation integrates **data-driven surrogates** into **PDES** to reduce simulation cost while preserving accuracy.
- A surrogate model predicts application behavior and decides when full PDES is needed.

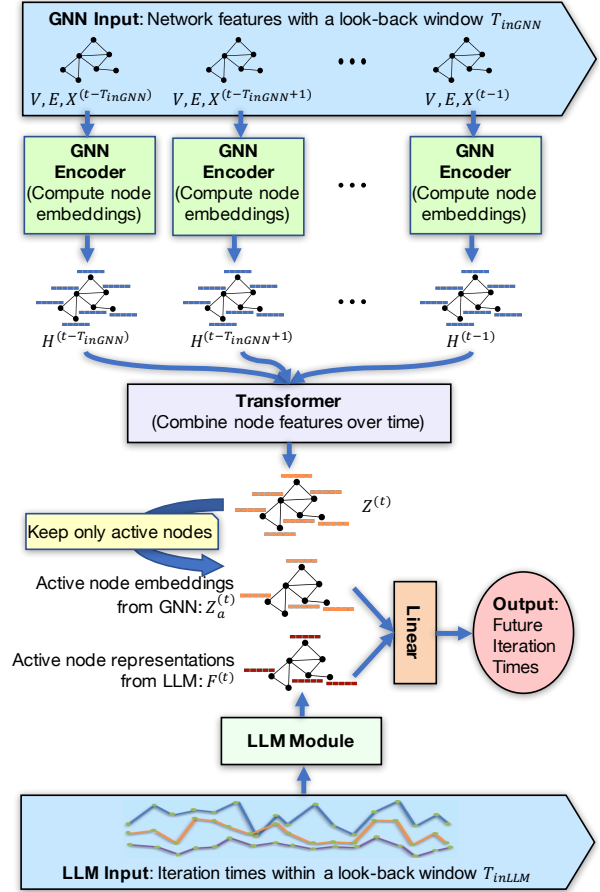
Key Idea

SMART is a lightweight surrogate model that combines:

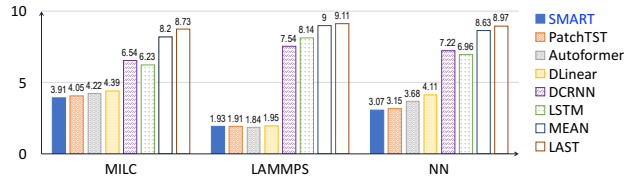
- GNNs to capture Dragonfly topology and **spatial contention**
- LLMs to model long-range **temporal patterns** in application runtime

Method, Results & Impact

SMART Architecture



Prediction Accuracy (MAPE)



Inference Efficiency (seconds/iteration)

Methods	Time per Iteration (s)
PDES	8.09 - 216.58
SMART	0.5150

Key Takeaways

- First domain-aware surrogate coupling GNNs + LLMs for Dragonfly
- Orders-of-magnitude speedup over high-fidelity PDES
- Accurate runtime prediction with less than 4% error